ICES Journal of Marine Science

ICES Journal of Marine Science (2020), 77(7-8), 2445–2455. doi:10.1093/icesjms/fsaa157

Food for Thought

Size-based theory for fisheries advice

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Andersen, K. H. Size-based theory for fisheries advice. – ICES Journal of Marine Science, 77: 2445–2455.

Received 16 April 2020; revised 7 August 2020; accepted 10 August 2020; advance access publication 13 November 2020.

Fisheries science and management is founded upon the Beverton–Holt theory of fish stock demography. The theory uses age as the structuring variable; however, there are several reasons to use body size as the structuring variable. Most of the processes that affect a fish are determined by its body size rather than its age: consumption, mortality, maturation, fecundity, fish gear selectivity, etc., and measurements of body size are easy, accurate, and abundant. Here, I review size-based theory of a fish stock and compare it to classic age-based theory. I show that size- and age-based demography are equivalent representations of demography. However, size-based theory is axiomatic, which leads to a deeper theory with two advantages: predictions need fewer parameters than age-based theory and the theory connects directly to life-history traits. The connection with traits makes size-based theory particularly useful for data-poor application and facilitates evolutionary calculations. I compare age- and size-based theories for fisheries impact and stock assessments and provide a perspective on the challenges and future of single-stock theory.

Keywords: Beverton–Holt, size based, size spectrum, traits

The increasing exploitation of fish stocks in the 20th century created a need for formal fisheries management. The scientific community responded by developing models of fish stock dynamics (e.g. Graham, 1935). Eventually, a consensus emerged as synthesized by Beverton and Holt (1957). Their age-based demographic description was built upon three assumptions: that growth is described by a von Bertalanffy equation, that natural mortality is constant, and that recruitment happens early in life. The theory has the advantage that it is simple and can easily be implemented numerically, initially in mechanical calculators and later in spreadsheets. The theory was good, and it has supported fisheries science and advice for more than half a century by answering relevant questions: what is the impact of fishing on different age groups in the fish stock? How much should we fish a stock? What are the relevant fisheries reference points (F_{msy}, F_{lim}, B_{msy}, etc.)? Finally, the theory formed the basis for heuristic and statistical fish stock assessment models. Fisheries advice without Beverton–Holt is unthinkable.

Today, new questions challenge age-based demography: how do we assess data-poor stocks where the parameters required for age-based assessments are not known? How do we assess the evolutionary side-effects of fishing (Jørgensen *et al.*, 2007)? How do we deal with density-dependent growth or cannibalism? The age-based framework may be adjusted to answer these questions; however, it does so with difficulty because it was only designed to predict demography. We need to reconsider our reliance on age-based demography as the theoretical basis of fisheries advice.

A fundamental shortcoming of age-based theory is that age is a weak predictor of many relevant processes. Body size—be it length or weight—has emerged as being more important than age for physiology (Winberg, 1956), mortality of individuals (McGurk, 1986), predator–prey interactions (Ursin, 1973), reproductive investment (Barneche *et al.*, 2018), fisheries impact through gear mesh regulation (Myers and Hoenig, 1997), and usually also market value. From biological, fisheries, and economic perspectives, body size is the natural structuring variable for fish demography.

In this Food for Thought, I review size-based theory of a fish stock and compare it to classic age-based demography. Broadly speaking, size-based theory has three advantages: (1) measuring

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size is easier, cheaper, and more accurate than measuring age; (2) it synthesizes four existing theories: size-based demography, life-history constants, evolutionary ecology and quantitative genetics, and trait-based ecology, into a coherent, axiomatic framework; (3) it opens up for new applications in data-poor stock assessment, assessment of fisheries-induced evolution, cross-species analyses, and assessment of emergent density dependence among adults. I will show how closely related the size- and age-based frameworks actually are; classic age-based Beverton–Holt is essentially a subset of the larger size-based theory. I will not dwell on the mathematical details, which are given elsewhere (Andersen, 2019), though I will compare the parameters between age- and size-based theories. I discuss pros and cons of size-based theory for fisheries impact and stock assessments, open issues, and the potential for future applications.

Emergence of size-based theory

Size-based demography has been a long time in the making. Jones (1974) developed the fundamental equations to calculate demography in length groups. At the same time, Andersen and Ursin (1977) developed their multi-species theory based on cohorts, but with all rates described by body mass. Beyer (1989) developed the first complete formulation of size-based demography of single fish stock. Despite size-based demography being available it had little impact as fisheries science focused on operationalizing existing age-based theory rather than developing new theory. The relative obscurity of size-based theory has led to confusion about its use and assumptions (Andersen *et al.*, 2016a) and errors in the implementation (Hordyk *et al.*, 2019).

Parallel to early size-based theory appeared the concept of life-history constants. Life-history constants are quantities that do not systematically vary between stocks or only have a weak taxonomic relation. The concept was pioneered by Beverton (1992) who found that the strong correlations between central parameters in age-based fish demography could be removed by formulating non-dimensional ratios. Examples are the M/K ratio or the ratio between size at maturation and asymptotic size. Life-history constants are part of general life-history theory developed through evolutionary ecology (Charnov, 1993).

Modern size-based theory combines the initial efforts of size-based demography with Beverton's life-history constants and evolutionary life-history theory into an internally consistent framework (Andersen, 2019). The cost of moving to a size-based framework is the loss of treasured concepts like the von Bertalanffy growth equation with the K and L_∞ parameters, abandoning the spreadsheet-friendly life tables, and scrapping the concepts of adult mortality, M and $M2$. Size-based theory replaces von Bertalanffy growth with physiology, life tables with partial differential equations, and the constant adult mortality with a size-based mortality. The absence of well-known concepts may seem overwhelming and makes the theory appear inaccessible. The reward is a theory that is consistently built upon a few fundamental assumptions and that can perform the same calculations as age-based demography, but that can also be used to estimate the evolutionary rates of life-history traits, be a basis for data-poor stock assessments or consumer-resource dynamics, and which provides general insights into fish life-history theory.

What is size-based demography?

Demography describes the size and structure of populations and their relation to growth, mortality, reproduction, etc. Classic age-based demography calculates numbers N_α at age α, usually under the assumption that the natural mortality M is constant (see Table A1). Weight-at-age is (usually) described with a von Bertalanffy growth formulation (Figure 1) or supplied from measurements. Multiplying numbers-at-age with weight-at-age gives cohort biomass, and summing all cohorts gives the spawning stock biomass. Reproduction is described by recruitment at an early age, which is calculated from a stock–recruitment relationship or considered constant for yield-per-recruit calculations. Age-based demography is specified by seven parameters: the von Bertalanffy growth parameters, K, L_∞, and t_0 (alternatively a weight-at-age table), the natural mortality M, age at maturation α_{mat}, and typically two parameters for the recruitment, for example the slope at origin and the maximum recruitment (Table 1).

Size-based demography is based on size groups instead of age groups. Mathematically, size-based demography is most conveniently represented as a continuum, but for numerical calculations (and here), it is represented by discrete size groups. Both length and weight groups can be used to represent body size, but weight groups lead to the cleanest theoretical formulation. Conversion between weight and length can be done superficially with the classic cube-law relation $w = 0.01l^3$ (with weight w in gram and length l in cm), or a more accurate formulation for the specific group or species (Froese, 2006). We need to decide upon which size groups to use. In age-based demography, 1-year age groups are the natural choice, but in size-based demography there is no such obvious grouping. The size range is typically evenly divided on a log scale from weight at recruitment w_R to the asymptotic weight W_∞ (Table A1), e.g. 2–4 g, 4–8 g. This procedure results in the complication that numbers-at-size depends on the size range of each group. The narrower the size range, the fewer fish in a size group. To deal with this complication, size-based demography is customarily described with the *size spectrum*—also referred to as the *normalized* size spectrum (Sprules and Barth, 2016)—$N(w)$, where w is the body mass (which can be substituted with length with appropriate conversion). The size spectrum is formed by dividing each size group by the size range it represents. The size spectrum is then a density function, similar to the probability density function from statistics, with unit numbers per body weight. While the probability density function is

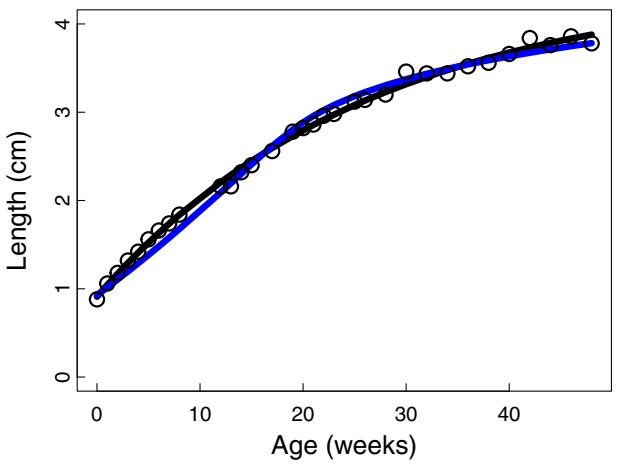


Figure 1. Length-at-age of a single female guppy (*Poecilia reticulata*) fitted to von Bertalanffy growth (black) and bi-phasic growth (grey). Data from Ursin (1967).

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normalized such that the integral over the function is 1, the size spectrum is normalized such that the integral (or sum of all groups each multiplied by its size range) is the total number of individuals in the population.

Figure 2 compares age- and size-based representations of demography in terms of numbers and biomass. Age-based demography clearly shows how abundance declines with age and that cohort biomass increases until around age of maturation (Figure 2a and c). The size spectra have a bump around size at maturation (Figure 2b and d). The bump appears because the growth rate declines around maturation and consequently individuals enter size groups at a slower rate than they leave them and pile up in the groups—the same mechanism that creates traffic jams on a highway. It is possible to convert between age- and size-based representations by multiplying with the growth rate: $N(\alpha) = N(w)g(w)\Delta\alpha$, where $\Delta\alpha = 1$ year is the age span in an age group. Biomass in size-based demography is conveniently represented by the *Sheldon spectrum*. The Sheldon spectrum is the biomass in logarithmically wide size groups, e.g. from 2 to 4 g, 4 to 8 g, 8 to 16 g. The Sheldon spectrum is found from the number spectrum by multiplying with weight^2 (Figure 2d). Each representation, age or size based, numbers, or biomass, has advantages and there is no universally "best" representation. They all represent the same demography.

Table 1. Parameters in age- and size-based representations.

Parameter	Age based	Size based
Asymptotic size	Free	Free
Growth	Free	$A = 5.35\ \mathrm{g}^{0.25}\ \mathrm{year}^{-1}$ [1 . . . 100]
Natural mortality[a]	Free	$a = 0.42$ [0.17 . . . 67]
Maturation	Free	$w_{\mathrm{mat}} = \eta_{\mathrm{mat}} W_\infty$; $\eta_{\mathrm{mat}} = 0.28$ [0.1 . . . 0.8]
Egg weight	–	0.001 g [0.0001 . . . 0.01]
Recruitment slope at origin[b]	Free	$\epsilon = 0.0066$ [0 . . . 1]
Max. recruitment	Free	Free

Ranges are given in brackets. "free" means that the parameter has to be specified on a per-stock basis. Values are from Andersen (2019).

[a]The mortality in size-based demography is specified as the dimensionless ratio between mortality and mass-specific rate of energy acquisition.

[b]The parameter ϵ can conveniently be divided into two parameters, the reproductive efficiency ϵ_{egg} and the recruitment efficiency ϵ_{R}, but for simplicity, they are here merged into $\epsilon = \epsilon_{\mathrm{R}}\epsilon_{\mathrm{egg}}$.

What is size-based theory?

A mathematical theory is based on a set of assumptions or axioms. Age-based demography is based on assumptions of a growth equation, a mortality (usually constant), and a recruitment formulation. From these assumptions follows predictions of fisheries

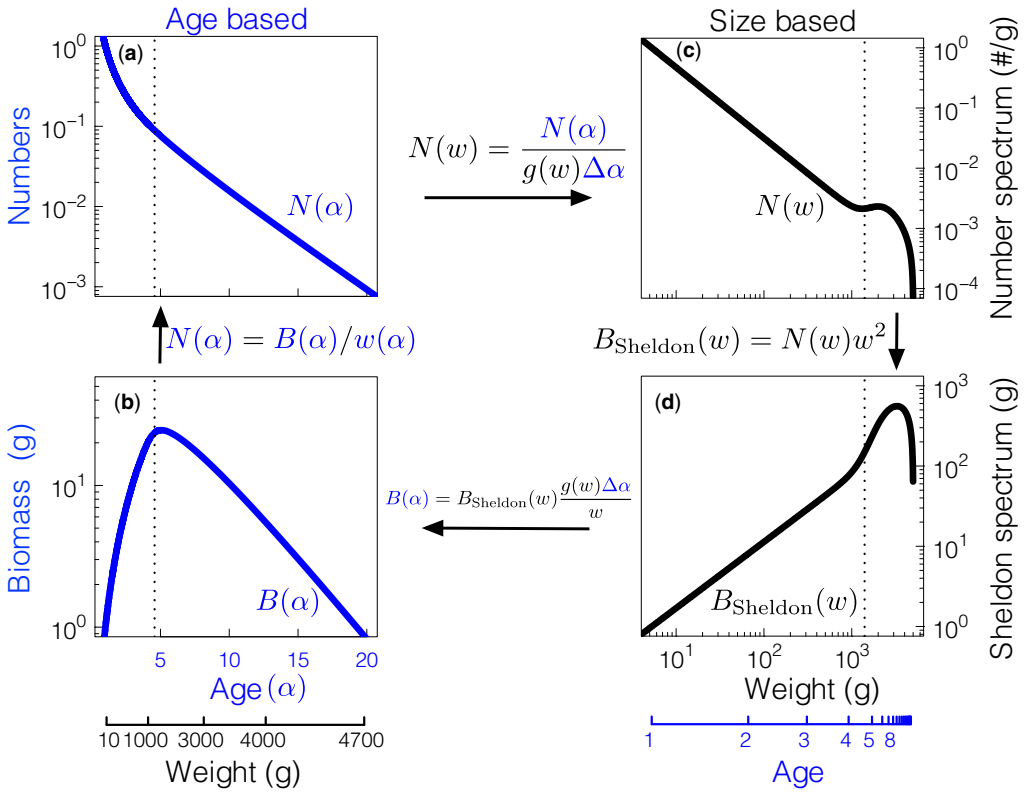


Figure 2. Age- and size-based representations of demography of a fish stock with asymptotic mass 5 kg. The first row (a+b) shows abundance as numbers-at-age $N(\alpha)$ (age α in years) and as the abundance size spectrum $N(w)$ (numbers per gram). The second row (b+d) shows cohort biomass $B(\alpha)$ and the Sheldon spectrum B_{Sheldon} (biomass in log-width bins; grams). All panels are normalized to 1 at recruitment (1 year; 5.3 g). The arrows show conversions between the four representations with $\Delta\alpha$ being the width of age groups (units of years) and $g(w)$ (gram per year) being the growth rate. The vertical dotted lines show size and age at maturation. The calculations are done with size-based theory, a bi-phasic growth equation, and parameters from Table 1. Derivation of the conversion between age- and size-based representations is given in Supplementary Appendix A.

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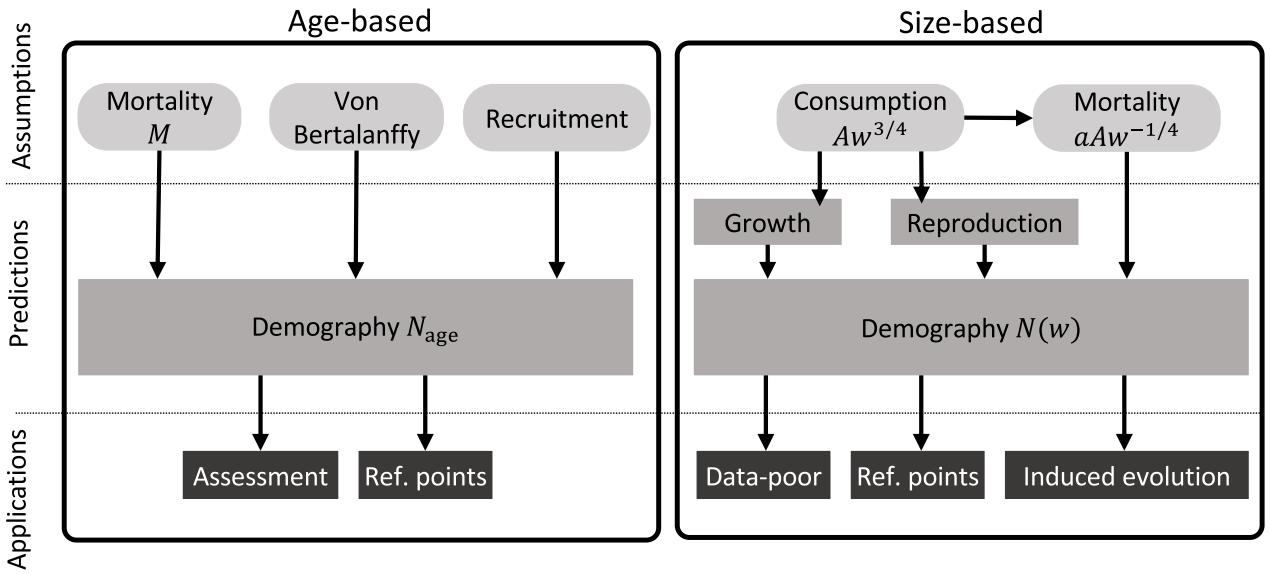


Figure 3. Overview of the assumptions, predictions, and applications of age- and size-based theories. Weight-at-age in age-based theory could either be represented with a von Bertalanffy growth curve or with measured weight-at-ages. Note the arrow from the consumption towards mortality, which illustrates the trade-off between available energy and mortality (assumption 2).

Table 2. Bi-phasic growth equations.

The von Bertalanffy growth equation is based on an energy budget of an individual fish with weight w:

$$g(w) = Aw^n - kw,$$

where A is a growth constant and k represents losses. Customarily the exponent $n = 2/3$ is used (but see Ursin, 1967). The growth constant has fractional units g^{n-1} per year. Fractional units are uncommon, but are perfectly valid. One should just think of A as the constant that scales the growth rate. Writing the growth rate as $g(w) = \mathrm{d}w/\mathrm{d}t$, the growth model can be solved to yield weight $w(\alpha)$ and length $l(\alpha)$ at age α:

$$w(\alpha) = W_\infty(1 - e^{-(1-n)AW_\infty^{n-1}\alpha})^{1/(1-n)} \quad \text{and} \quad l(\alpha) = L_\infty(1 - e^{-K\alpha}).$$

The solution of $w(\alpha)$ is valid for asymptotic sizes $W_\infty \gg w_{egg}$. The length-based solution introduces the von Bertalanffy growth constant K, which is derived from A and L_∞ (Andersen *et al.*, 2009, see also Table 1).
Bi-phasic growth models are based on the same energy budget (Roff, 1983; Lester *et al.*, 2004; Quince *et al.*, 2008). They assume that the losses, kw, represent expenses for reproduction including migration, forgone feeding, and the actual egg or gonad production. These losses therefore mainly occur when the individual is mature. Maturation is described with a maturity function $\psi_{mat}(w)$ that switches smoothly between 0 and 1 around the size at maturation:

$$g(w) = Aw^n - \psi_{mat}(w)kw,$$

where $n = 3/4$ is customarily used. The differences between solutions of the von Bertalanffy growth model (with $n = 2/3$) and the bi-phasic growth model (with $n = 3/4$) are fairly small (Fig. 1). This formulation has the advantage that it predicts the number of eggs produced by an individual R_p:

$$R_p(w) = \epsilon_{egg}\psi_{mat}(w)kw/w_{egg} = \epsilon_{egg}\psi_{mat}(w)AW_\infty^{n-1}w/w_{egg},$$

where ϵ_{egg} is the fraction of the reproductive investment that results in eggs. Integrating over the entire stock structure gives the total reproductive output, which predicts the slope at origin in the recruitment function (Andersen and Beyer, 2015, see also Table 1).

yield and reference points, F_{msy}, F_{lim}, etc. (Figure 3a). Size-based theory is based on three similar assumptions:

Assumption 1: *The available energy for growth and reproduction* (*mass per time*) *scales with body mass as* $Aw^{3/4}$. This assumption is in line with the standard von Bertalanffy growth equation, though the exponent is usually set to 3/4 instead of 2/3 (see Table 2). The 3/4 exponent lines up with modern metabolic theory (West *et al.*, 1997) and has some empirical support (Essington *et al.*, 2001). However, whether 3/4 or 2/3 is used

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does not make much of a difference and the assumption that available energy scales with body weight is essentially the same between age- and size-based theories.

Assumption 2: *Natural mortality declines with size and is proportional to available energy*: $\mu(w) = aAw^{-1/4}$. The decline of mortality with a $-1/4$ exponent with body mass has some empirical support (McGurk, 1986; Hirst and Kiørboe, 2002). The proportionality with available energy means that faster-growing fish also have a higher mortality. In classic life-history theory, this corresponds to a constant "M/K" ratio. The relation between available energy and mortality also has empirical support in relation to growth (e.g. Lankford *et al.*, 2001). Both declining mortality and the relation between available energy and mortality follow from size spectrum theory: if consumption scales as in assumption 1, and bigger organisms eat smaller ones, then mortality scales as assumption 2 (Andersen and Beyer, 2006; Andersen *et al.*, 2009). This means that size-based theory is inherently a metabolic theory at the level of individuals; the only essential assumption is that metabolism and consumption increases as a power-law function with mass (assumption 1). The constant a is called the *physiological mortality* (Beyer, 1989) and is a central parameter in size-based theory. It represents the growth:mortality ratio in same way as the M/K is in age-based demography, though based on a declining mortality with size.

Assumption 3: *Density dependence operates early in life, before fishing starts and before maturation.* This assumption means that growth and mortality are independent of the abundance of the stock; there is no density-dependent growth and mortality. The assumption works well when the stock is depleted by fishing (Andersen *et al.*, 2016b) but may be less appropriate for unfished or lightly fished stocks where density-dependent reductions in adult growth or cannibalism may occur (Zimmermann *et al.*, 2018)—I will come back to this later. This assumption is the same between age- and size-based theories.

From assumption 1 about available energy follows growth and reproductive output as a bi-phasic growth equation (Table 2; Figure 3b). The bi-phasic growth equation is very similar to the von Bertalanffy growth equation and with the quality of extant data practically indistinguishable. Bi-phasic growth has a kink around the size at maturation where energy begins to be allocated from growth towards reproduction.

Size-based theory is not strict about any of the assumptions. One can use another growth function than the one following from assumption 1 or another mortality than assumption 2 and still apply the theory—just as age-based theory can use other growth equations than von Bertalanffy and non-constant mortality. Sticking to the formulations in assumptions 1 and 2, however, makes it possible to rely on constant life-history parameters and formulate the trade-offs needed for quantitative genetics. In the final section, I discuss how to relax the third assumption and operate with density-dependent growth and reproduction happening later in life.

Size- vs. age-based theory and applications

From the three assumptions follows predictions of growth, the slope of the recruitment function, and demography (Figure 3b). As it appears from Table A1, the equations have a fairly similar structure between age- and size-based formulations (at least in the discrete formulation used here). However, the parameters are different. Instead of the von Bertalanffy K parameter, size-based theory uses the growth constant A; instead of the constant natural mortality M, it uses the physiological mortality a; instead of age at maturation α_{mat}, it specifies the size at maturation as proportional to the asymptotic weight $\eta_{mat} W_\infty$; instead of specifying recruitment slope at origin (or the steepness), it operates with a "recruitment efficiency" ϵ. The age- and size-based parameters are connected and one can convert between them, though not perfectly in all cases (Table A2).

There are smaller differences between the demographic theories. For example, size-based theory operates with a size-dependent mortality whereas age-based theory typically operates with a constant mortality. This, however, is a small difference and age-based demography can easily operate with an age-dependent mortality. The growth equations are also slightly different; age-based theory uses a von Bertalanffy growth equation, whereas size-based theory operates with a bi-phasic growth equation. Nevertheless, they are fairly similar, and again, age-based theory could just as well use a bi-phasic growth equation (or vice versa). We can therefore conclude that age- and size-based demography are essentially identical. They are two equally valid representations of fish demography.

So, when should one use one framework over the other? In the following, I discuss pros and cons of each framework in terms of the theoretic formulations (1–3), applications to stock assessments and other fisheries advice (4–6), and the use of size/age as a structuring variable (7–9):

(1) Size-based theory provides a prediction of egg production. Hence, it does not need an external stock-specific parameter to characterize recruitment (the slope at origin or the steepness). The prediction emerges from the bio-energetic budget (Table 2). This idea has been re-invented many times (Andersen and Ursin, 1977; Roff, 1983; Lester *et al.*, 2004; Mangel *et al.*, 2010; Andersen and Beyer, 2015) and is central in physiologically structured models (de Roos and Persson, 2013) as well as in evolutionary theory, where accounting for the cost of reproduction is essential (Day and Taylor, 1997). The prediction of recruitment from size-based theory can of course be used in age-based demographic calculations.

(2) The parameters in size-based theory are for the most part life-history constants. This means that they are not strongly correlated with other parameters and they have an average expected value across all fish stocks (Table 1). Instead of a von Bertalanffy growth constant K that is correlated with asymptotic size, size based uses the growth constant A, which does not, or only weakly, correlate with asymptotic size (Figure 4). The growth parameter A directly indicates somatic growth rate; if A is larger/smaller than about $5\,g^{0.25}$ year^{-1} fish in the stock grows faster/slower than average. Von Bertalanffy K states the maturation rate—how fast maturity is approached—which does not indicate growth rate directly. Similarly, the relative maturation size η_{mat} can be compared across stocks, while the age at maturation varies with a combination of K and L_∞. The recruitment parameter—slope at origin—is one of the big unknowns in fish demography, and it varies systematically with asymptotic size (Hall *et al.*, 2006; Thorson, 2020). In contrast, the recruitment efficiency ϵ has a similar value across stocks—even though it is still uncertain with a large variability. Formulating theory in terms of

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life-history constants is the ultimate realization of Ray Beverton's project about life-history constants in fish stocks (Beverton, 1992). The relations between the size-based life-history constants and the classic age-based parameters are given in Table A2.

(3) Size-based theory can be interpreted as a trait-based theory because of the constant life-history parameters. The "master trait" is the asymptotic size W_∞, and we can make general predictions across all species just knowing the asymptotic size. Trait-based predictions reveal an important insight: population-level rates, such as fisheries reference points and maximum population growth rates, do not scale metabolically with asymptotic size as $W_\infty^{-1/4}$ (Figure 5a and b; Andersen and Beyer, 2015). In other words, smaller species do not necessarily have higher population growth rates than large species. This is a surprising finding because it clashes with metabolic theory, despite size-based theory being based on metabolic assumptions at the level of individuals. The finding emerges because bony fish make eggs with approximately the same size, irrespective of their asymptotic size (Neuheimer *et al.*, 2015). That metabolic scaling rules do not apply for fish has important implications for simple food web calculations or unstructured theory that often rely on metabolic scaling rules for parameterization of P/B rates or population growth rates. While it is clear that asymptotic size is the most important trait, it is less clear which trait to include next. It depends on the application. . For the example with the maximum population growth rate r_{max} and F_{msy} shown here, it is the growth rate A (Figure 4); if the application is an exploration of the importance of big fish for reproduction, the ratio between asymptotic size and size at maturation η_{mat} is the most relevant trait (Andersen *et al.*, 2019); if the application is stock assessment, it is the growth:-mortality ratio a (Kokkalis *et al.*, 2015). As the variation in the life-history parameters is known (Table 1), the uncertainty of prediction based on just one trait—e.g. asymptotic size—can be represented by a Monte Carlo simulation with random values of the other life-history parameters within their known ranges (see for example Andersen and Brander, 2009; Andersen and Beyer, 2015). Alternatively, empirical relations between taxonomy and life-history parameters (Thorson *et al.*, 2017)—e.g. a (M/K) or η_{mat}—can be used to further refine the estimates.

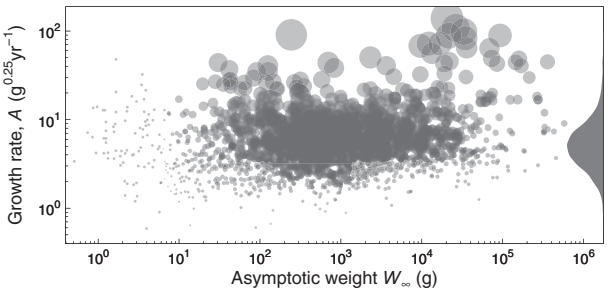


Figure 4. The trait space of bony fish (Actinopterygii) as defined by their asymptotic weight, W_∞, and their growth rate constant, A. The area of each point is proportional to the maximum population growth rate as estimated from W_∞, A, and the default life-history parameters in Table 1 (calculation in Supplementary Appendix C). The grey patch to the right shows the distribution of A with a geometric mean of 5.15 $g^{1/4}$ $year^{-1}$. Data are from FishBase (Froese and Pauly, 2013) selected with the criterion that t_0 is in the range ± 1. Note that because there is no correction for taxonomy the figure has an over-representation of species like cod, rockfish, etc.A more proper representation could be found by applying FishLife (Thorson et al, 2020).

Size-based theory has applications for stock assessments and impact assessments, both in data-rich and data-poor cases, and for evolutionary impact assessments:

(4) Fish stock assessment is widely done with age-structured models (or production models). Age-based assessment relies on aging of samples from surveys and catches, a procedure which is costly, only resolves age on a yearly basis, and is prone to error. Size-based assessments use measurements of length or weight that are cheap, precise, and with little scope for error of interpretation. A size assessment essentially estimates the log–log slope of the size distribution (Figure 2a), which is directly related to the mortality: growth ratio a; the

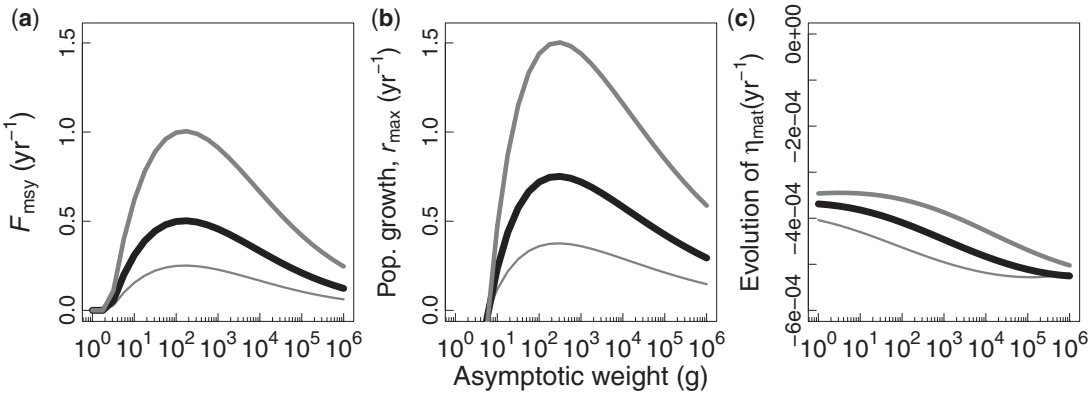


Figure 5. Trait-based predictions from size-based theory of (a) F_{msy}, (b) maximum population growth rate r_{max}, and (c) fisheries-induced evolution of size at maturation relative to weight at maturation. Each prediction is made for slow- to fast-growing species (thin/thick grey lines correspond to half/double the average growth rate). The prediction of r_{max} is based on the approximation in Supplementary Appendix 7, which underestimates r_{max} for small species. The predictions of F_{msy} and evolutionary responses are made with a trawl selectivity with a 50% selectivity at 5% of the maximum weight and a fishing mortality of 0.3 $year^{-1}$.

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steeper the slope, the higher the mortality or the slower the growth. However, to know the mortality—and then the fishing mortality—we need to multiply the slope with the growth rate constant A. In other words, we need to know the growth rate, which requires aging. Some of the advantages of the size-based assessment therefore disappears because if we have to do aging we could just as well do age-based assessments. Some information of growth rate can be found by following cohorts in size (e.g. Fournier *et al.*, 1998), but this is fairly unreliable in many cases. However, if we are satisfied with only knowing the status of the stock, i.e. the ratio between fishing mortality and the reference point, for example F/F_{msy}, then we do not need to know growth. The reason being that most reference points are also proportional to the growth rate A, and therefore, the ratio F/F_{msy} becomes independent of the growth rate (Kokkalis *et al.*, 2015). If we are willing to sacrifice the accurate estimation of F and just estimate F/F_{msy}, then size-based assessment will be as accurate as age-based assessment, even without knowledge of growth (Kokkalis *et al.*, 2017). As many central management decisions are made with reference to the stock status F/F_{msy} (or similar ratios with other reference points), knowledge of the absolute fishing mortality is of secondary relevance and there is a big untapped potential for size-based assessment of stock status.

(5) The central role of life-history constants and traits makes the theory the natural starting point for data-poor stock assessments and impact assessments. Size-based data-poor stock assessment methods are actively being developed (e.g. Le Quesne and Jennings, 2012; Hordyk *et al.*, 2015; Kokkalis *et al.*, 2017) (though at times they fail to properly account for the difference between age- and size-based demography; Hordyk *et al.*, 2019). They all rely on some variant of life-history constants as a "Robin Hood" method of using knowledge from other stocks to determine parameter values. In the extreme case, knowing just the asymptotic size we can estimate the fisheries reference points, even accounting for density-dependent recruitment. As explained above, the main unknown, the growth rate A, disappears if we form the ratio of observed fishing mortality and the fisheries reference point. This means that we can predict the status of the fishery, whether it is over- or under-exploited with respect to F_{msy} (Kokkalis *et al.*, 2015). We can even account for uncertainty by varying the life-history parameters within their known ranges (Kokkalis *et al.*, 2017). Knowledge of some parameters, for example the growth parameter A, the size at maturation, the growth:mortality ratio, can be used to refine the predictions, e.g. from taxonomically informed methods (Thorson, 2020). Size-based theory should be the starting point for size-based data-poor stock assessment.

(6) The full bio-energetic formulation with a trade-off between consumption and mortality (assumption 1) and between somatic growth and reproduction (Table 2) opens the door to quantitative genetics-based predictions of fisheries-induced evolution (Figure 5c; Andersen and Brander, 2009). Size-based theory can therefore be used to make evolutionary impact assessment of fishing. The example given here shows how size (and age) of maturation declines with increased fishing mortality. A similar pattern emerges if natural mortality is increased, in accordance with observations of a correlation between M/K and size of maturation (Thorson *et al.*, 2017).

There are also some advantages of an age-based formulation of demography:

(7) Age-based demography is easier to implement quickly (compare the equations in Table A1). However, as age-based demography is done with age groups of 1 year we need to compensate for the mortality during the year with Baranov's catch equation when calculating yield. This correction is not needed in size-based theory, which typically operates with many (100s) of size groups. Age-based representations are also more intuitive; from Figure 2a and c, we immediately see the number and the biomass of fish with a certain age. The size-based representation shows the number density, or the Sheldon biomass spectrum, which are more abstract representations of demography. There is a trade-off between the simplicity of age-based and the higher resolution of size-based demography.

(8) Age-based theory naturally incorporates the annual schedule that governs many aspects of fish demography, in particular, annual reproduction. Resolving an annual schedule in size-based theory does not come naturally.

(9) While most relevant physiological processes and gear selectivity depend more on size than on age, one effect does depend on age: senescence. There is little evidence for senescent drop in reproduction, but some evidence for senescent mortality (Beverton *et al.*, 2004). However, senescence is unlikely to be relevant in fished populations, where older individuals are very rare, so it is safe to ignore senescence for most applications to fished populations. If needed, age-based senescence can be resolved in size-based theory with a size-age relation, just like size-based mortality can be incorporated in age-based theory.

Overall, the difference between age- and size-based theories is in the scope of the theoretical framework. Size-based theory is a comprehensive axiomatic framework within which age-based demography is embedded as a special case. Predictions from size-based theory of parameters, recruitment, etc., can be used for age-based demography. The choice between age- and size-based demography is therefore often one of the conveniences: if annual schedules are central, an age-based formulation is to be preferred; if precision of resolution is important, or data-poor or fisheries-induced evolution applications is needed, then size based is preferred. For stock assessments, in particular, in data-poor situations, there are potential big gains in moving to a size-based formulation because of the easy access to cheap and accurate size measurements.

Challenges and future perspectives

Size-based methods have maturated to be useful for practical fisheries advice, and it is already actively used to develop data-poor stock assessments (e.g. Hordyk *et al.*, 2015; Kokkalis *et al.*, 2017),

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in ecosystem models (Scott *et al.*, 2014; Spence *et al.*, 2020), and for assessments of fisheries-induced evolution (Andersen and Brander, 2009). This situation should not keep us from testing and developing the theory further. Three areas warrant attention: the bio-energetic basis, predictions of recruitment, and applications in consumer-resource modelling for modelling-dependent growth and mortality.

The core of size-based theory is the bio-energetic budget used to calculate growth and reproduction (Table 2). Fairly little experimental work on bio-energetic growth models has been done since Ursin (1967), probably due to the difficulty and expense of working experimentally with long-lived species like fish. The consequence is that we have an incomplete understanding of exactly which processes determine investment in reproduction, metabolism, and asymptotic size, and how they are related. There are evident trade-offs between these three processes, for example higher metabolic costs and activity leads to smaller asymptotic size everything else being equal. However, everything else is not equal; reproduction also plays a role and how the organism balances this three-way trade-off is unknown. Furthermore, the reproductive output is often not just proportional to mass but in many cases increases faster (Barneche *et al.*, 2018). Is this process due to ecology—size-based variability in available food and body condition as seen for Icelandic cod (Marteinsdottir and Begg, 2002)—or is it an evolutionary adaptation? Knowing whether more than proportional investment in reproduction is due to the environment or is a species trait matters for statistical estimation, which often assume a taxonomic signal. A similar problem is how bio-energetic processes depend on temperature, where our current understanding has been thrown into doubt (Lefevre *et al.*, 2017; Jutfelt *et al.*, 2018). What is missing is not more theory but long-term experiments on individual fish, carefully designed to inform about the trade-offs between growth, metabolism, activity, reproduction, maturation, and asymptotic size, and how they all respond to temperature, both for short-term acclimation, longer-term adaptation, within and across species of different asymptotic sizes.

The prediction that the recruitment slope at origin declines with asymptotic size and increases with growth rate (Table A1) is crucial because it determines how fisheries reference points scale with asymptotic size (Andersen and Beyer, 2015). The prediction that slope decrease with asymptotic size conforms qualitatively with analyses of the RAM database (Hall *et al.*, 2006; Thorson, 2020). However, the observed slope declines faster than predicted (roughly an exponent −0.76 vs. predicted −0.25). The predicted scale of the slope as $W_\infty^{-1/4}$ leads to fisheries references points that only differ slightly between small and large species (Figure 5b). In contrast, if the slope is ≈ -0.93, close to the observed −0.76, then fisheries reference points scale metabolically. That means that the F_{msy} of a 5 g forage fish is 10 times higher than that of a 50-kg large demersal species (see Supplementary Appendix C). However, observations of F_{msy} show that F_{msy} is roughly similar between small and large species (Andersen and Beyer, 2015; Jacobsen, 2015). The importance of the prediction of the recruitment slope for fisheries reference points makes it an important question to resolve. Working with recruitment data is difficult because of the very noisy nature of the observations and the difficulty in assessing the slope at origin or steepness from the

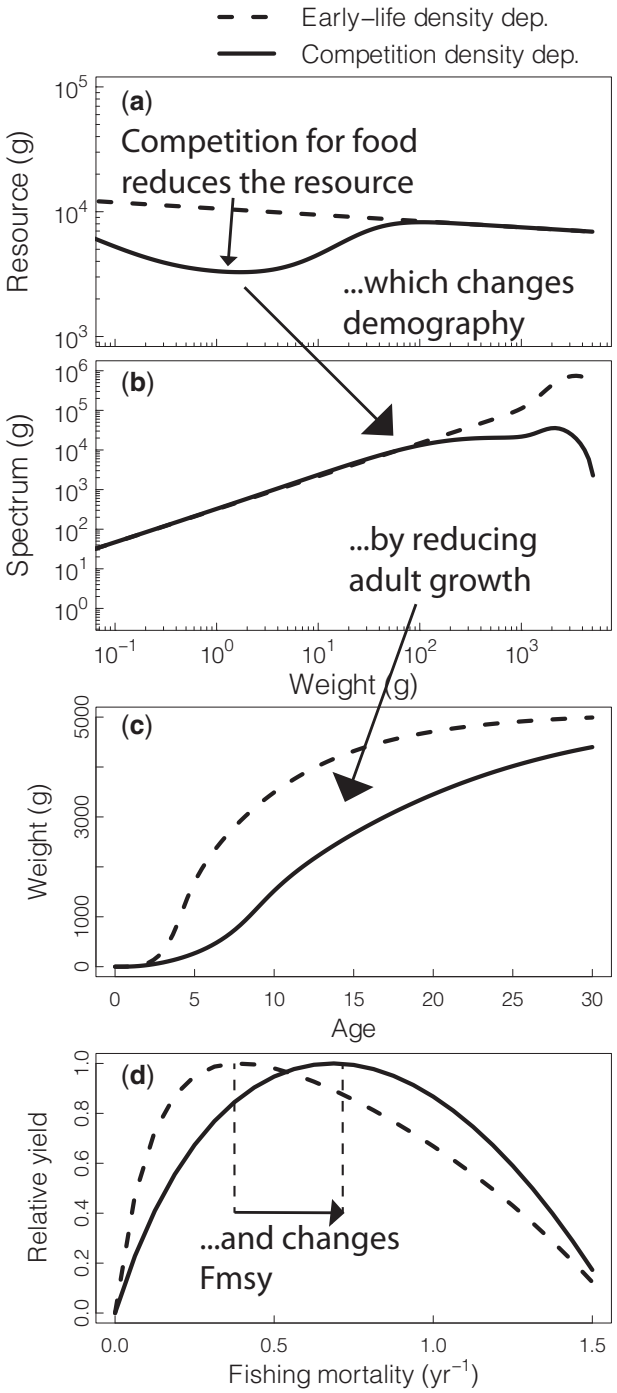


Figure 6. Comparison between a full consumer-resource calculation (solid lines) and classic theory with early-life density dependence (dashed lines). The resource represents an even size distribution of prey (a). Competition for food by adults reduces the resource. The reduced resource leads to slower growth (c) and a reduced adult abundance (b). The density dependence due to competition may change the reference points (d). Here, the F_{msy} is increased but the change depends on details of the size selection by fishing (Andersen *et al.*, 2016b) and the relative strength of early and competition density dependence (van Gemert and Andersen, 2018b). The model is described in Andersen (2019, chap. 10).

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observations. Furthermore, the data need to be corrected for age/size at recruitment, which implies further assumptions.

Finally, there is an increasing realization of the importance of accounting for density-dependent growth (Lorenzen and Enberg, 2002; Zimmermann *et al.*, 2018) and cannibalism in the impact assessments of fisheries (van Gemert and Andersen, 2018a). Density-dependent processes that occur later in life break assumption 3, which is a foundation of both age- and size-based demography. Density-dependent growth and cannibalism can potentially change reference points fundamentally (Andersen *et al.*, 2016b) and induce Allee effects that hinder recovery (Van Leeuwen *et al.*, 2008). Even though it is still unclear how important such effects are for reference points (van Gemert and Andersen, 2018b), it is likely that fisheries science and management will increasingly be asked to assess density-dependent processes besides stock–recruitment.

Resolving density-dependent growth and mortality can be done by formulating heuristic relations between growth (A) and mortality (a) and stock biomass, such that growth declines (or mortality increases) as the stock increases (Lorenzen, 2008; van Gemert and Andersen, 2018b). A deeper understanding requires a consumer-resource perspective, such as is commonly taken in ecology. Physiologically structured modelling is the natural approach to combine classic consumer-resource theory with structured populations in a fisheries context (Persson *et al.*, 2014). In physiologically structured models, all density dependence emerges from competition for a resource or from cannibalism (de Roos and Persson, 2013). Size-based theory is naturally extended to a physiologically structured model, however, with the slight difference that it also incorporates a stock–recruitment relation to account for early-life density dependence due to habitat limitation or larvae survival (Andersen, 2019, chap. 10) or spatial population dynamics (Andersen *et al.*, 2016b; see Figure 6 for an example of the mechanism in a size-based consumer-resource model and the implications for F_{msy}). It is an open question whether these more advanced consumer-resource models can be integrated into practical fisheries advice: do we have the knowledge to parameterize them, and the ability to apply the predictions for management?

Conclusion

The emergence of size-based theory does not replace existing age-based theory. Rather, it embraces and expands it. Size-based theory offers a solid axiomatic framework with more applications than age-based theory and is directly linked to modern ecological and evolutionary theories. Instead of having to develop disjoint frameworks for data-poor assessments, fisheries-induced evolution, density dependent processes, etc., size-based theory offers a unified framework. The theory is formulated with specific assumptions about the growth equation, about mortality, and about reproductive investments. It is important to remember that size-based theory is not dogmatic with regards to these formulations; another growth equation, mortality, or formulation of reproductive investment can be used if it is more appropriate—as can also be done in age-based demography. However, the simple formulation offers a route to make impact assessment with little data and to make sweeping cross-taxa prediction. Finally, size-based demography links to directly to size-based models of the entire fish community that are being increasingly used as models of intermediate complexity (Scott *et al.*, 2014; Spence *et al.*, 2020). Despite the advantages of size-based theory, age-based demography will continue to form the basis of much fisheries advice because it is well known and well established. However, where we develop new advice for emerging fish stocks or in countries with poorly developed management, it should be size based.

Supplementary data

Supplementary material is available at the *ICESJMS* online version of the manuscript.

Data availability

All calculations are done with the R package FishSizeSpectrum https://github.com/Kenhasteandersen/FishSizeSpectrum. Data and code for figures are available on GitHub: https://github.com/Kenhasteandersen/FishDemography.

Acknowledgements

Many thanks to Jan Beyer and Keith Brander for comments on a draft of the manuscript. The work is supported by the VKR Foundation through the Ocean Life Center of Excellence and by the PANDORA EU H2020 project.

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